Uniform Penrose stability from boundary exclusion including the directional low-frequency response

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doi:10.5802/jep.353; CC BY4.0

Editorial note. The parameter convention delta_0 in(0,1] is retained as the exact source356 fragment. It is the positive regularity slack needed by the directional Cauchy–Schwarz estimate and uniform tail bound. The unrelated stronger nonlinear-theorem Sobolev assumptions on source356 and radial sufficient cases near212 are not imported.

Editorial note. The low-frequency limiting response L_0(mu,omega) is an explicit hypothesis. It is not the point value L(lambda,0;hbar)=0, and cannot be dropped when applying compactness.

Editorial note. The proof itself gives the bound for hbar in[0,1], including the defined classical limit; the original uniform Penrose definition requests hbar in(0,1].

Editorial note. Derivatives of widehat(g) in the directional trace estimate mean Fourier-argument derivatives evaluated on the indicated line, as also fixed by the chain-rule factor in1122. No derivative of the normalized direction map is being estimated.

Editorial note. No assumption of monotone/radial equilibrium from the following proposition is needed: the counted hypotheses are the original explicit real/imaginary boundary conditions.

Editorial scope. One uniform stability criterion with its full local Fourier-direction and response estimates. The following smallness/radial sufficient cases1192-1254 are not counted as extra tasks; full Hartree damping is also not claimed.

Allowed standard prerequisites (editorial).

- Complex argument principle and dominated convergence. Exclude zeros for each fixed parameter and establish continuity of the compactified response; pointwise zero-freeness is not assumed to give a uniform gap.
- Fourier Plancherel and Riemann-Lebesgue lemmas. Move between moment/Sobolev control and Fourier traces, and control large spatial frequencies.
- Sobolev line trace and polynomial-weight ordering. Ordinary Sobolev infrastructure; the directional weighted L1 estimate1037-1056 and all response bounds remain proved obligations.

for some constant $\delta_0 \in (0,1]$

Definition 0.1 (Uniform Penrose condition)**.** *An interaction kernel $w\colon \mathbb{R}^d \to \mathbb{R}$ and steady state $\gamma_g = (2\pi\hbar)^d g(-i\hbar\nabla)$ satisfy the uniform Penrose condition if there exists a constant $c_{\mathrm{P}} > 0$ such that the inequality*

$$\inf_{\substack{\operatorname{Re}\lambda \geq 0 \\ k \in \mathbb{R}^d}} \left| 1 + \widetilde{\mathscr{L}}(\lambda, k) \right| \geq c_{\mathrm{P}} \tag{1}$$

holds uniformly for all $\hbar \in (0,1]$, where

$$\widetilde{\mathscr{L}}(\lambda, k) = \frac{2}{\hbar}\, \widehat{w}(k) \int_0^\infty e^{-\lambda t} \sin\left(\tfrac{1}{2}\hbar t |k|^2\right) \widehat{g}\,(kt)\; dt.$$

0.1 Quantum operators and norms

To formulate phase-mixing and scattering estimates in the quantum setting, quantum analogues of classical derivatives and weighted Sobolev norms are introduced. The standard and double Japanese brackets are defined by $\langle x \rangle := (1 + |x|^2)^{1/2}$ and $\langle x, y \rangle := (1 + |x|^2 + |y|^2)^{1/2}$, respectively. For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}^d$, write $|\alpha| = \alpha_1 + \cdots + \alpha_d$ and $D^\alpha = \partial_1^{\alpha_1} \cdots \partial_d^{\alpha_d}$.

The classical weighted Sobolev norm $\|\cdot\|_{H^\sigma_M}$ is now recalled.

Definition 0.2 (Weighted Sobolev norm)**.** *Let $\sigma \geq 0$ and let $M \geq 0$ be an integer. For a function $g = g(v)$, the weighted Sobolev norm H^σ_M is defined as*

$$\|g\|_{H^\sigma_M} := \sum_{|\alpha| \leq M} \|v^\alpha \langle \nabla \rangle^\sigma g\|_{L^2}.$$

For a function $f = f(x, v)$, the weighted Sobolev norm H^σ_M is defined as

$$\|f\|_{H^\sigma_M} := \sum_{|\alpha| \leq M} \|v^\alpha \langle \nabla_x, \nabla_v \rangle^\sigma f\|_{L^2},$$

where the weight applies only to the velocity variable.

Throughout, integrals such as $\int_x dx$ denote integration over $\mathbb{R}^d$ in the x variable. The following conventions are adopted for the Fourier transform of functions $g = g(v)$ and $f = f(x, v)$:

$$\widehat{g}(p) := \int_v e^{-ip\cdot v} g(v)\, dv, \qquad \widehat{f}(k, p) := \int_x \int_v e^{-ik\cdot x} e^{-ip\cdot v} f(x, v)\, dv dx.$$

The following lemma shows that, in the space H^σ_M, moments and derivatives can be reordered freely, allowing them to be applied in whichever order is most convenient.

Lemma 0.3 (Ordering of moments)**.** *Let $\sigma \geq 0$, let $M \in \mathbb{N}$, and $g \in H^\sigma_M(\mathbb{R}^d)$. Then the following norms are equivalent:*

$$\|g\|_{H^\sigma_M} := \sum_{|\alpha| \leq M} \|\langle \nabla_k \rangle^\sigma (k^\alpha g)\|_{L^2} \sim \sum_{|\alpha| \leq M} \|k^\alpha \langle \nabla_k \rangle^\sigma g\|_{L^2}.$$

The Sobolev L^2 trace lemma is also recalled.

Lemma 0.4 (L^2 trace [1, Lem. 2.15])**.** *Let $g \in H^s(\mathbb{R}^d)$ with $s > {}^{(d-1)}\!/_2$ and $C \subset \mathbb{R}^d$ be an arbitrary straight line. Then, there holds*

$$\|g\|_{L^2(C)} \lesssim_{s,d} \|g\|_{H^s(\mathbb{R}^d)}.$$

$$\widetilde{\mathscr{L}}(\lambda, k; \hbar) = \frac{2}{\hbar|k|}\widehat{w}(k) \int_0^\infty e^{-\lambda s/|k|} \sin(\tfrac{\hbar}{2}|k|s)\widehat{g}\big(\tfrac{k}{|k|}s\big)\, ds. \tag{2}$$

To compare with the classical setting, consider the limit

$$\widetilde{\mathscr{L}}(\lambda, k; 0) := \lim_{\hbar \to 0} \widetilde{\mathscr{L}}(\lambda, k; \hbar),$$

which evaluates to

$$\widetilde{\mathscr{L}}(\lambda, k; 0) = \widehat{w}(k) \int_0^\infty e^{-\lambda s/|k|} s\widehat{g}\big(\tfrac{k}{|k|}s\big)\, ds.$$

Next, using the argument principle, a condition on the behaviour of $\widetilde{\mathscr{L}}$ along the imaginary axis is shown to imply the uniform Penrose condition. The low frequencies require separate attention: $\widetilde{\mathscr{L}}(\lambda, 0; \hbar) = 0$, whereas (2) and dominated convergence give, for every $\hbar \in [0, 1]$, $\operatorname{Re}\mu \geq 0$ and $\omega \in \mathbb{S}^{d-1}$,

$$\lim_{r \downarrow 0} \widetilde{\mathscr{L}}(r\mu, r\omega; \hbar) = \widetilde{\mathscr{L}_0}(\mu, \omega) := \widehat{w}(0) \int_0^\infty e^{-\mu s} s\, \widehat{g}(\omega s)\, ds,$$

so $\widetilde{\mathscr{L}}$ is not in general continuous at $(\lambda, k) = (0, 0)$.

Lemma 0.5. *Suppose $w \in L^1(\mathbb{R}^d)$, $g \in H^{3/2+\delta_0}_{1+\lceil d/2 \rceil}(\mathbb{R}^d)$, and that the following implications hold:*

$$\forall \hbar \in [0, 1],\, \tau \in \mathbb{R},\, k \in \mathbb{R}^d, \quad \operatorname{Im}\widetilde{\mathscr{L}}(i\tau, k; \hbar) = 0 \implies \operatorname{Re}\widetilde{\mathscr{L}}(i\tau, k; \hbar) > -1, \tag{3}$$

and, for the low-frequency response,

$$\forall \tau \in \mathbb{R},\, \omega \in \mathbb{S}^{d-1}, \quad \operatorname{Im}\widetilde{\mathscr{L}_0}(i\tau, \omega) = 0 \implies \operatorname{Re}\widetilde{\mathscr{L}_0}(i\tau, \omega) > -1. \tag{4}$$

Then the uniform Penrose condition (1) *is satisfied.*

These lemmas imply the following estimate for directional integrals of derivatives of $\widehat{g}$:

Lemma 0.6. *Let $n \geq 0$, and let α be a multi-index. Then, for every $k \neq 0$,*

$$\int_0^\infty \langle s\rangle^n \big|\partial_k^\alpha \widehat{g}\big(\tfrac{k}{|k|}s\big)\big|\, ds \lesssim_{d,\delta_0,n,|\alpha|} \|g\|_{H^{n+1/2+\delta_0}_{|\alpha|+\lceil d/2 \rceil}}.$$

Proof. Applying Cauchy–Schwarz followed by Lemma 0.4,

$$\begin{aligned}\int_0^\infty \langle s\rangle^n \big|\partial_k^\alpha \widehat{g}\big(\tfrac{k}{|k|}s\big)\big|\, ds &\lesssim_{\delta_0} \left(\int_0^\infty \langle s\rangle^{2n+1+2\delta_0} \big|\partial_k^\alpha \widehat{g}\big(\tfrac{k}{|k|}s\big)\big|^2\, ds\right)^{1/2} \\ &\lesssim_d \left(\int_{\mathbb{R}^d_k} \Big|\langle \nabla_k\rangle^{\lceil d/2 \rceil} \big(\langle k\rangle^{n+1/2+\delta_0} \partial_k^\alpha \widehat{g}(k)\big)\Big|^2\, dk\right)^{1/2}.\end{aligned}$$

Then, by Plancherel and Lemma 0.3:

$$\begin{aligned}\int_0^\infty \langle s\rangle^n \big|\partial_k^\alpha \widehat{g}\big(\tfrac{k}{|k|}s\big)\big|\,ds &\lesssim_d \big\|\langle x\rangle^{\lceil d/2\rceil}\langle\nabla_x\rangle^{n+1/2+\delta_0}\,(x^\alpha g(x))\big\|_{L^2}\\ &\lesssim \sum_{|\beta|\le\lceil d/2\rceil}\big\|x^\beta\langle\nabla_x\rangle^{n+1/2+\delta_0}\,(x^\alpha g(x))\big\|_{L^2}\\ &\lesssim_n \sum_{|\beta|\le\lceil d/2\rceil}\big\|\langle\nabla_x\rangle^{n+1/2+\delta_0}\,\big(x^{\beta+\alpha}g(x)\big)\big\|_{L^2}\\ &\lesssim_\alpha \|g\|_{H^{n+1/2+\delta_0}_{|\alpha|+\lceil d/2\rceil}}.\end{aligned}$$

□

Lemma 0.7. *Suppose $w\in L^1(\mathbb{R}^d)$ and $g\in H^{3/2+\delta_0}_{\lceil d/2\rceil}(\mathbb{R}^d)$. Then:*

a. *As $|k|\to\infty$, the quantity $\widetilde{\mathscr{L}}(\lambda,k;\hbar)\to 0$, with decay uniform in $\operatorname{Re}\lambda\ge 0$ and $\hbar\in[0,1]$.*

b. *There exists a constant $C_1=C_1(d,\delta_0)$, independent of $\hbar\in[0,1]$, such that for all $\operatorname{Re}\lambda\ge 0$ and $k\in\mathbb{R}^d$,*

$$\big|\widetilde{\mathscr{L}}(\lambda,k;\hbar)\big|\le C_1\|w\|_{L^1}\|g\|_{H^{3/2+\delta_0}_{\lceil d/2\rceil}}.$$

c. *If in addition $g\in H^{3/2+\delta_0}_{1+\lceil d/2\rceil}(\mathbb{R}^d)$, then there exists a constant $C_2=C_2(d,\delta_0)$, independent of $\hbar\in[0,1]$, such that for all $\operatorname{Re}\lambda\ge 0$ with $\lambda\neq 0$ and $k\in\mathbb{R}^d$,*

$$\big|\widetilde{\mathscr{L}}(\lambda,k;\hbar)\big|\le C_2\frac{|k|}{|\lambda|}\|w\|_{L^1}\|g\|_{H^{3/2+\delta_0}_{1+\lceil d/2\rceil}}.$$

Proof. First observe that $\widetilde{\mathscr{L}}(\lambda,k=0;\hbar)=0$ for all $\operatorname{Re}\lambda\ge 0$ and $\hbar\in[0,1]$. Now fix $\hbar\in(0,1]$. For $k\neq 0$ and $\operatorname{Re}\lambda\ge 0$, the representation (2) together with Lemma 0.6 yields the estimate

$$\begin{aligned}\big|\widetilde{\mathscr{L}}(\lambda,k;\hbar)\big| &\lesssim_d |\widehat{w}(k)|\frac{1}{\hbar|k|}\int_0^\infty \big|\sin\big(\tfrac{\hbar}{2}|k|s\big)\;\widehat{g}\big(\tfrac{k}{|k|}s\big)\big|\,ds\\ &\lesssim |\widehat{w}(k)|\int_0^\infty s\,\big|\widehat{g}\big(\tfrac{k}{|k|}s\big)\big|\,ds\\ &\lesssim_{d,\delta_0} |\widehat{w}(k)|\|g\|_{H^{3/2+\delta_0}_{\lceil d/2\rceil}}.\end{aligned}$$

Since $w\in L^1(\mathbb{R}^d)$, the Fourier transform $\widehat{w}(k)\to 0$ as $|k|\to\infty$, establishing part (a). Moreover, $|\widehat{w}(k)|\le\|w\|_{L^1}$ for all k, yielding part (b).

For part (c), the expression (2) can be integrated by parts in s, with no boundary contributions, to obtain

$$\begin{aligned}&\widetilde{\mathscr{L}}(\lambda,k;\hbar)\\ &\quad= 2\widehat{w}(k)\frac{1}{\hbar|k|}\Big(-\frac{|k|}{\lambda}\Big)\int_0^\infty \partial_s\big(e^{-\lambda s/|k|}\big)\sin\big(\tfrac{\hbar}{2}|k|s\big)\,\widehat{g}\big(\tfrac{k}{|k|}s\big)\,ds\\ &\quad= 2\widehat{w}(k)\frac{1}{\hbar\lambda}\int_0^\infty e^{-\lambda s/|k|}\Big[\tfrac{\hbar}{2}|k|\cos\big(\tfrac{\hbar}{2}|k|s\big)\widehat{g}\big(\tfrac{k}{|k|}s\big)+\sin\big(\tfrac{\hbar}{2}|k|s\big)\tfrac{k}{|k|}\cdot\nabla_k\widehat{g}\big(\tfrac{k}{|k|}s\big)\Big]\,ds.\end{aligned}$$

Therefore,

$$\begin{aligned}\left|\widetilde{\mathscr{L}}(\lambda,k;\hbar)\right| &\lesssim \frac{|k|}{|\lambda|}\|w\|_{L^1}\int_0^\infty \Big[\big|\widehat{g}\big(\tfrac{k}{|k|}s\big)\big| + s\big|\nabla_k \widehat{g}\big(\tfrac{k}{|k|}s\big)\big|\Big]\,ds \\ &\lesssim_{d,\delta_0} \frac{|k|}{|\lambda|}\|w\|_{L^1}\|g\|_{H^{3/2+\delta_0}_{1+\lceil d/2\rceil}}.\end{aligned}$$

The same bounds hold for the classical limit $\widetilde{\mathscr{L}}(\lambda,k;0)$, with identical estimates. □

Proof. For each $\hbar\in[0,1]$ and $k\in\mathbb{R}^d$, the assumptions on w and g ensure that the map $\lambda\mapsto\widetilde{\mathscr{L}}(\lambda,k;\hbar)$ is analytic in the half-plane $\{\operatorname{Re}\lambda>0\}$, continuous on $\{\operatorname{Re}\lambda\geq 0\}$ and, by Lemma 0.7 (c), tends to zero as $|\lambda|\to\infty$. By the argument principle, the number of zeros of the function $\lambda\mapsto 1+\widetilde{\mathscr{L}}(\lambda,k;\hbar)$ in $\{\operatorname{Re}\lambda>0\}$ is equal to the winding number of the curve

$$\Gamma := \big\{\widetilde{\mathscr{L}}(i\tau,k;\hbar) : \tau\in\mathbb{R}\big\}$$

around the point -1. The implication (3) ensures that Γ does not meet the half-line $(-\infty,-1]$. It follows that $1+\widetilde{\mathscr{L}}(\lambda,k;\hbar)$ has no zeros in $\{\operatorname{Re}\lambda\geq 0\}$. The same argument, applied to $\mu\mapsto 1+\widetilde{\mathscr{L}_0}(\mu,\omega)$ for fixed $\omega\in\mathbb{S}^{d-1}$ and using (4), shows that $1+\widetilde{\mathscr{L}_0}(\mu,\omega)$ has no zeros in $\{\operatorname{Re}\mu\geq 0\}$.

It remains to show that the infimum in (1) is positive. For $k\neq 0$, $\hbar\in[0,1]$ and $\operatorname{Re}\lambda\geq 0$, the representation (2) reads

$$\begin{aligned}\widetilde{\mathscr{L}}(\lambda,k;\hbar) &= \widehat{w}(k)\,\Phi\left({}^{\lambda}\!/_{|k|},\hbar|k|,{}^{k}\!/_{|k|}\right),\\ \Phi(\mu,h,\omega) &:= \int_0^\infty e^{-\mu s}\,\frac{2}{h}\sin\left({}^{hs}\!/_{2}\right)\widehat{g}(\omega s)\,ds,\end{aligned}$$

where the factor $2h^{-1}\sin(hs/2)$ is understood as s when $h=0$, so that $\widetilde{\mathscr{L}_0}(\mu,\omega)=\widehat{w}(0)\Phi(\mu,0,\omega)$. The function Φ is continuous on $\{\operatorname{Re}\mu\geq 0\}\times[0,\infty)\times\mathbb{S}^{d-1}$. Indeed, the integrand is continuous and bounded by $s\|\widehat{g}\|_{L^\infty}$, so the integral over $[0,R]$ is continuous for every $R>0$, while $\left|2h^{-1}\sin(hs/2)\right|\leq s$ and the Cauchy–Schwarz and trace estimates in the proof of Lemma 0.6 give, uniformly in $\operatorname{Re}\mu\geq 0$, $h\geq 0$ and $\omega\in\mathbb{S}^{d-1}$,

$$\begin{aligned}\int_R^\infty \left|e^{-\mu s}\,\frac{2}{h}\sin\left({}^{hs}\!/_{2}\right)\widehat{g}(\omega s)\right|ds &\leq \left(\int_R^\infty \frac{s^2}{\langle s\rangle^{3+2\delta_0}}\,ds\right)^{1/2}\left(\int_0^\infty \langle s\rangle^{3+2\delta_0}|\widehat{g}(\omega s)|^2\,ds\right)^{1/2}\\ &\lesssim_{d,\delta_0} R^{-\delta_0}\|g\|_{H^{3/2+\delta_0}_{\lceil d/2\rceil}},\end{aligned}$$

so Φ is a uniform limit of continuous functions. Now set

$$F(\mu,r,\omega,\hbar) := 1+\widehat{w}(r\omega)\,\Phi(\mu,\hbar r,\omega),$$

so that $F(\mu,r,\omega,\hbar)=1+\widetilde{\mathscr{L}}(r\mu,r\omega;\hbar)$ for $r>0$ and $F(\mu,0,\omega,\hbar)=1+\widetilde{\mathscr{L}_0}(\mu,\omega)$.

By Lemma 0.7, there exist constants $K>0$ and $M>0$ such that, for $k\neq 0$ and $\operatorname{Re}\lambda\geq 0$, if either (i) $|k|\geq K$, or (ii) $|\lambda|\geq M|k|$, then

$$\left|1+\widetilde{\mathscr{L}}(\lambda,k;\hbar)\right|\geq\frac{1}{2}$$

uniformly in $\hbar \in [0,1]$, while $\left|1 + \widetilde{\mathscr{L}}(\lambda, 0; \hbar)\right| = 1$. Writing $\lambda = r\mu$ and $k = r\omega$, the remaining region corresponds to the compact set

$$\mathcal{K} := \left\{(\mu, r, \omega, \hbar) : \operatorname{Re}\mu \geq 0,\, |\mu| \leq M,\, 0 \leq r \leq K,\, \omega \in \mathbb{S}^{d-1},\, \hbar \in [0,1]\right\}.$$

On this compact set, the function F is continuous, since $\widehat{w}$ and Φ are, and it does not vanish, by the first paragraph. Hence

$$c_{\mathrm{P}} := \min\left\{1/2,\, \min_{\mathcal{K}} |F|\right\} > 0$$

is a lower bound for $\left|1 + \widetilde{\mathscr{L}}(\lambda, k; \hbar)\right|$ over $\operatorname{Re}\lambda \geq 0$, $k \in \mathbb{R}^d$ and $\hbar \in [0,1]$. This establishes the uniform Penrose condition (1). □

References

[1] J. Bedrossian, N. Masmoudi & C. Mouhot - "Landau damping in finite regularity for unconfined systems with screened interactions", Comm. Pure Appl. Math. 71 (2018) no. 3, p. 537-576